

- 6 Y is a discrete random variable which takes the values $0, 2, 4, \dots$. The probability generating function of Y is given by

$$G_Y(t) = \frac{k}{1 - at^2}.$$

- (a) Find k in terms of a . [1]

[illegible]

- (b)** Show that $P(Y > 2) = a^2$. [3]

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(c) Find the value of $E(Y)$.

[2]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dashed lines, creating a series of uniform gaps for letter height. The lines are evenly spaced across the entire page, providing a guide for consistent letter formation. There is no text or other markings on the page.